Calculus II

Solution to Assignment 1

1. (a)

$$\ln\frac{2}{1}+\ln\frac{3}{2}+\ln\frac{4}{3}+\ln\frac{5}{4}+\cdots=\sum_{n=1}^{\infty}\ln(\frac{n+1}{n})=\sum_{n=1}^{\infty}[\ln(n+1)-\ln n]$$

Since $S_k=\sum_{n=1}^{k}[\ln(n+1)-\ln n]=\ln 2+(\ln 3-\ln 2)+(\ln 4-\ln 3)+(\ln 5-\ln 4)+\cdots+(\ln(k+1)-\ln k)=\ln(k+1)$ diverges as $k\to\infty$, this series diverges.

(b)

$$\frac{2}{3}+\frac{2}{3^2}+\frac{2}{3^3}+\frac{2}{3^4}+\cdots=\sum_{n=1}^{\infty}\frac{2}{3^n}$$

This is a geometric series with the first term $a=\frac{2}{3}$ and common ratio $r=\frac{1}{3}$. So the series converges and its sum is $S=\frac{a}{1-r}=\frac{2/3}{1-1/3}=1$.

(c)

$$\left(\frac{1}{2}+\frac{1}{3}\right)+\left(\frac{1}{2^2}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}+\frac{1}{3^3}\right)+\cdots=\sum_{n=1}^{\infty}\left(\frac{1}{2^n}+\frac{1}{3^n}\right)=\sum_{n=1}^{\infty}\frac{1}{2^n}+\sum_{n=1}^{\infty}\frac{1}{3^n}$$

Both $\sum_{n=1}^{\infty}\frac{1}{2^n}$ and $\sum_{n=1}^{\infty}\frac{1}{3^n}$ are geometric series. Refer to (b), we can get $\sum_{n=1}^{\infty}\frac{1}{2^n}=1$ and $\sum_{n=1}^{\infty}\frac{1}{3^n}=\frac{1}{2}$. So the sum of the asked series is $\frac{3}{2}$.

(d)

$$\frac{1}{2}+\frac{3}{4}+\frac{5}{6}+\frac{7}{8}+\cdots=\sum_{n=1}^{\infty}\left(\frac{2n-1}{2n}\right)=\sum_{n=1}^{\infty}\left(1-\frac{1}{2n}\right)$$

Since $\lim_{n\to\infty}1-\frac{1}{2n}=1\neq 0$, the series is divergent.

2. (a)

$$\sum_{n=1}^{\infty}(\sqrt{n+1}-\sqrt{n})=(\sqrt{2}-\sqrt{1})+(\sqrt{3}-\sqrt{2})+(\sqrt{4}-\sqrt{3})+\cdots=\lim_{n\to\infty}\sqrt{n+1}-1=\infty$$

Therefore, the series diverges.

(b)

$$\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\cdots+\frac{1}{(2n-1)(2n+1)}+\cdots=\sum_{n=1}^{\infty}\frac{1}{(2n-1)(2n+1)}=\frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)$$

This is another telescoping series. When expanded, most terms cancel out:

$$\frac{1}{2}\left((1-\frac{1}{3})+(\frac{1}{3}-\frac{1}{5})+(\frac{1}{5}-\frac{1}{7})+\cdots\right)$$

All intermediate terms cancel, leaving:

$$\frac{1}{2}(1 - \lim_{n\to\infty} \frac{1}{2n+1}) = \frac{1}{2}$$

Thus, the series converges and its sum is $\frac{1}{2}$.

(c) Since the limit of $\sin \frac{n\pi}{6}$ as $n \to \infty$ does not exist, the series is divergent.

3. Any series of the form

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

where $0 < p \leq 1$.

4. Multiply both sides by $n \cdot (n+1) \cdot (n+2)$ to clear the denominators:

$$1 = a(n+1)(n+2) + b(n)(n+2) + c(n)(n+1)$$

$$1 = (a+b+c)n^2 + (3a+2b+c)n + 2a$$

For the equation to hold for all n, the coefficients of n^2, n, and the constant term must be equal on both sides. This gives us the system of equations:

$$a + b + c = 0$$

$$3a + 2b + c = 0$$

$$2a = 1$$

Solving these equations, we find:

$$a = \frac{1}{2}, \quad b = -1, \quad c = \frac{1}{2}$$

Thus, the series can be written as:

$$\sum_{n=1}^{\infty} \left(\frac{1/2}{n} - \frac{1}{n+1} + \frac{1/2}{n+2} \right)$$

This series is telescoping. Most terms cancel out, leaving:

$$\frac{1/2}{2} + \lim_{n\to\infty} \left(-\frac{1/2}{n+1} + \frac{1/2}{n+2} \right) = \frac{1}{4}$$

Thus, the sum of the series is $\frac{1}{4}$.

5. (a) The series $-\frac{8}{9}+\frac{8^2}{9^2}-\frac{8^3}{9^3}+\cdots$ is a geometric series with $a=-\frac{8}{9}$ and $r=-\frac{8}{9}$. It converges with sum:

$$S=\frac{-\frac{8}{9}}{1-(-\frac{8}{9})}=-\frac{8}{17}$$

(b) The series $\frac{1}{3}+\frac{1}{6}+\frac{1}{9}+\cdots=\frac{1}{3}\sum_{n=1}^{\infty}\frac{1}{n}$, which diverges.

(c) The series $\frac{1}{3}+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt[3]{3}}+\cdots$ diverges because $\lim_{n\to\infty}\frac{1}{\sqrt[n]{3}}=1\neq 0$.

(d) The series $\frac{3}{2}+\frac{3^2}{2^2}+\frac{3^3}{2^3}+\cdots$ is a geometric series with $|r|>1$, so it diverges.

6. (a) Since $5+n^2>n^2$ for all $n\in\mathbb{N}$, the series $\frac{1}{5+1^2}+\frac{1}{5+2^2}+\frac{1}{5+3^2}+\cdots=\sum_{n=1}^{\infty}\frac{1}{5+n^2}$ converges by comparison with $\sum_{n=1}^{\infty}\frac{1}{n^2}$.

(b) The series $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n(n^4+1)}}$ converges as

$$\lim_{n\to\infty}\frac{1/n^2}{1/\sqrt{n(n^4+1)}}=\lim_{n\to\infty}\sqrt{n+\frac{1}{n^3}}=\infty$$

and $\sum_{n=1}^{\infty}\frac{1}{n^2}$ converges.

(c) If $a=1$, the series $\sum_{n=1}^{\infty}\frac{1}{1+a^n}=\sum_{n=1}^{\infty}\frac{1}{2}=\infty$.
If $a<1$, $\lim_{n\to\infty}\frac{1}{1+a^n}=1\neq 0$. So it diverges.
If $a>1$, $\sum_{n=1}^{\infty}\frac{1}{1+a^n}$ converges by comparison with $\sum_{n=1}^{\infty}\frac{1}{a^n}$.

(d) Since

$$\lim_{n\to\infty}\frac{\ln(1+n)}{n}=\lim_{x\to\infty}\frac{\ln(1+x)}{x}=\lim_{x\to\infty}\frac{1/(1+x)}{1}=0$$

the series $\sum_{n=1}^{\infty}\frac{1}{\ln(1+n)}$ diverges by the comparison with $\sum_{n=1}^{\infty}\frac{1}{n}$.

(e) The series $\sum_{n=1}^{\infty}\sin\frac{1}{n}$ diverges as $\lim_{n\to\infty}\frac{\sin\frac{1}{n}}{1/n}=1$ and $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges.

(f) Since $0<r<1$ and $\cos^2 n\theta$ is bounded between 0 and 1, each term $r^n\cos^2 n\theta$ is less than or equal to r^n. The series $\sum_{n=0}^{\infty}r^n$ converges. By the comparison test, the given series also converges.

(g) $1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+\frac{1}{9}+\cdots=\sum_{n=1}^{\infty}\frac{1}{2n-1}$. Since $\lim_{n\to\infty}\frac{1/n}{1/(2n-1)}=2$ and $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges, the series $\sum_{n=1}^{\infty}\frac{1}{2n-1}$ also diverges.

(h) If $x=0$, $\sum_{n=1}^{\infty}\left(1-\cos\frac{x}{n}\right)=0$ converges. If $x\neq 0$,

$$\lim_{n\to\infty}\frac{1/n^2}{1-\cos(x/n)}\overset{y=1/n}{=}\lim_{y\to 0}\frac{y^2}{1-\cos(xy)}=\lim_{y\to 0}\frac{2y}{x\sin(xy)}=\lim_{y\to 0}\frac{2}{x^2\cos(xy)}=\frac{2}{x^2}$$

Also because $\sum_{n=1}^{\infty}\frac{1}{n^2}$ converges, the series $\sum_{n=1}^{\infty}\left(1-\cos\frac{x}{n}\right)$ also converges for $x\neq 0$.

(i) Since

$$\lim_{n\to\infty} \frac{1/n}{\ln\left(1+\frac{x}{n}\right)} \overset{y=1/n}{=} \lim_{y\to 0} \frac{y}{\ln(1+yx)} = \lim_{y\to 0} \frac{1}{1/(1+yx)\cdot x} = \frac{1}{x}$$

and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, the series $\sum_{n=1}^{\infty} \ln\left(1+\frac{x}{n}\right)$ also diverges.